

Chapter 4 Warm-Up: Boolean Logic and Decisions

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1 Boolean Variables

1. A Boolean variable can have one of two values: `True` or `False`. Comparisons also produce Boolean values.

- Enter the following statements in the interpreter:

```
is_raining = False
temperature = 72

print(is_raining)
print(temperature > 70)
print(temperature == 72)
print(temperature < 60)
```

2 Boolean Operations and Decisions

1. Python combines Boolean conditions using `and`, `or`, and `not`.

- Predict the values of

```
print(True and True)
print(True and False)
print(True or False)
print(False or False)
print(not True)
```

- Now write a short program that collects information about a student:

```
score = int(input("Score: "))

submitted = input(
    "Submitted? yes/no: ") == "yes"

attendance = input(
    "Attendance OK? yes/no: ") == "yes"

Create the Boolean variable

ready = submitted and attendance

and use it in the following chained decision:

if score >= 90 and ready:
    print("Excellent")
elif score >= 70 and ready:
    print("Pass")
elif not submitted or not attendance:
    print("Incomplete")
else:
    print("Revise")
```

Run the program several times. Find inputs that cause each of the four branches to execute.

3 Truth-Table Exercise

1. Associativity changes the grouping of operands, while commutativity changes their order. We will test whether these changes affect a Boolean expression. Consider the four expressions

```
E1 = (p and q) and (not r)
E2 = p and (q and (not r))
E3 = (not r) and (q and p)
E4 = p and q and not r
```

Here, **E2** changes the association of the `and` operations, **E3** changes both grouping and ordering, and **E4** relies on Python's precedence rules.

- There are eight possible combinations of three Boolean variables. Complete the truth table:

<i>p</i>	<i>q</i>	<i>r</i>	E1	E2	E3	E4
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

- Check the table using Python. For each row, assign values to `p`, `q`, and `r`, calculate all four expressions, and print the result.
- Compare the four result columns. Does changing the **association** of the `and` operations change the truth value?
- Does changing the **order** of the Boolean variables change the truth value?
- Based on your truth table, what evidence do you have that Boolean `and` is associative and commutative?